

main.c

Run

Output

```
1 #include <stdio.h>
2 int main()
3 {
4     int pages[50], frames[10];
5     int n, f, i, j, k;
6     int pageFault = 0;
7     int index = 0;
8     int found;
9     printf("Enter number of pages: ");
10    scanf("%d", &n);
11    printf("Enter page reference string:\n");
12    for(i = 0; i < n; i++)
13    {
14        scanf("%d", &pages[i]);
15    }
16    printf("Enter number of frames: ");
17    scanf("%d", &f);
18    for(i = 0; i < f; i++)
19    {
20        frames[i] = -1;
21    }
22    printf("\nPage\tFrames\n");
23    for(i = 0; i < n; i++)
24    {
25        found = 0;
26        for(j = 0; j < f; j++)
27        {
28            if(frames[j] == pages[i])
29            {
30                found = 1;
31                break;
32            }
33        }
34        if(found == 0)
35        {
36            frames[index] = pages[i];
37            index = (index + 1) % f;
38            pageFault++;
39        }
40        printf("%d\t", pages[i]);
41        for(k = 0; k < f; k++)
42        {
43            if(frames[k] != -1)
44                printf("%d ", frames[k]);
45            else
46                printf("- ");
47        }
48        if(found)
49            printf("Hit");
50        else
51            printf("Fault");
52        printf("\n");
53    }
54    printf("\nTotal Page Faults = %d\n", pageFault);
55    return 0;
```

Enter number of pages: 10
Enter page reference string:
1 2 3 1 4 5 2 1 2 3
Enter number of frames: 3

Page	Frames
1	1 - - Fault
2	1 2 - Fault
3	1 2 3 Fault
1	1 2 3 Hit
4	4 2 3 Fault
5	4 5 3 Fault
2	4 5 2 Fault
1	1 5 2 Fault
2	1 5 2 Hit
3	1 3 2 Fault

Total Page Faults = 8

=== Code Execution Successful ===